

SCIENCE

Simple Circuits — At-Home Activity

A safe, easy experiment your child can do at home to explore how electrical circuits work.

A safe, simple experiment your child can do at home to see how electricity flows through a complete circuit.

■ ■ Safety note

Use only a low-voltage battery (AA or D, 1.5V) as listed below. Never experiment with a wall outlet or household wiring.

What You Need

- 1 AA or D battery (1.5V)
- 1 small light bulb (flashlight bulb) or LED
- 2 pieces of insulated wire, about 15 cm each, with ends stripped
- Electrical tape (optional, to hold connections)

Steps

- 1 Look at the battery and find the flat end (–) and the bump end (+).
- 2 Connect one wire from the (+) end of the battery to one side of the bulb.
- 3 Connect the second wire from the other side of the bulb to the (–) end of the battery.
- 4 If the bulb doesn't light up, check that both wire ends are making solid metal-to-metal contact.
- 5 Once it works, try disconnecting one wire — notice the bulb turns off. This is an “open circuit.”
- 6 Reconnect it to make a “closed circuit” again and the bulb lights up.

What's Happening?

Electricity needs a complete loop — a “closed circuit” — to flow from the battery, through the wire and bulb, and back to the battery. If there's any gap in the loop (an “open circuit”), the electricity can't flow and the bulb stays off. This is the same idea behind light switches in your house!